

HARSHAVARDHAN S

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SUMMARY

Computer Science undergraduate (AI & Data Science) and a Data Analytics intern at Roche, focused on building AI systems end to end — from fine-tuning and serving domain-specific LLMs (PEFT, vLLM) to engineering and deploying full-stack applications that reach real users. Shipped a self-hosted, AI-powered nutrition app serving 20+ active users on self-managed GPU infrastructure. Seeking to build intelligent systems that go from prototype to production.

TECHNICAL SKILLS

- **AI / ML:** LLM Fine-tuning (PEFT, DoRA), vLLM, Hugging Face, PyTorch, NLP, Computer Vision, LangChain
- **Languages & Frameworks:** Python, FastAPI, Django, REST APIs, JavaScript, TypeScript, React
- **Tools & Fundamentals:** Docker, nginx, Linux, Git, SQLite; Data Structures & Algorithms, DBMS, OS

EXPERIENCE

Data Analytics Intern — Roche

Jun 2026 – Present

- Analyze clinical trial datasets to uncover patterns and trends that drive actionable insights for study teams.
- Clean, transform, and explore large trial datasets to support data quality and reliable reporting.
- Translate analytical findings into clear summaries that inform data-driven decisions.

PROJECTS

Domain-Specific Code LLM for a Proprietary Enterprise Platform

- Fine-tuned base models across multiple training paradigms — fill-in-the-middle (FIM), causal language modeling (CLM), supervised fine-tuning (SFT), and DoRA — to build a domain-specific code LLM for a proprietary enterprise development language, served at high throughput via vLLM behind an OpenWebUI interface.
- Reverse-engineered and rebuilt the platform's core, including an exact clone of its database engine (multi-value data model, maker-checker lifecycle, write-ahead-log crash recovery) exposed via REST API to generate and validate training data.
- Built a full-stack coding agent to automate code conversion into the platform's language, scaling dataset creation for fine-tuning.
- **Tools:** Python, PyTorch, Hugging Face, PEFT/DoRA, vLLM, OpenWebUI, FastAPI, Linux

Harahachi — Self-Hosted AI Calorie & Macro Tracker (20+ active users)

- Built a photo-first nutrition tracker where a vision LLM identifies each food component and estimates calories and macros; serving 20+ active users in production.
- Engineered a FastAPI edge-AI gateway with QR device pairing, multi-user accounts, a multi-LLM backend registry with automatic failover, and an admin dashboard.
- Deployed as a single Docker container (nginx + Cloudflare tunnel, outbound-only) on an NVIDIA DGX server; shipped as a PWA + Capacitor Android APK backed by a 150+ test suite.
- **Tools:** Python, FastAPI, SQLite, Docker, nginx, Cloudflare Tunnel, JavaScript, Capacitor

EDUCATION

B.Tech — Computer Science and Engineering (AI & Data Science)

Expected 2027

SASTRA Deemed to be University, Thanjavur | CGPA: 8.25

Honors — Applied Accelerated Artificial Intelligence

NPTEL, IIT Guwahati

- GPU compute capabilities and system software for accelerated-computing AI solutions with end-to-end industrial deployment. Completed with Elite certification.

Honors — Cloud Computing

NPTEL, IIT Kharagpur

- Cloud architectures, virtualization, and service models; finished in the top 2% of 33,000+ certified candidates (Elite – Topper, 90%).

ADDITIONAL INFORMATION

- **Achievements:** Finalist, ANRF AISEHack 2026 (in coordination with IBM Research & IIT Delhi), held at IIIT Hyderabad — Team Hackaholics.
- **Certifications:** Hugging Face LLM Course (LLM Fundamentals; Fine-tuning Language Models); MathWorks ML & Deep Learning Onramp; AlgoUniversity Graph Theory Camp & DSA Course.
- **Languages:** English, Tamil, Hindi (understanding proficiency).